EUCLID III.16 / SAVED EXPERIMENT

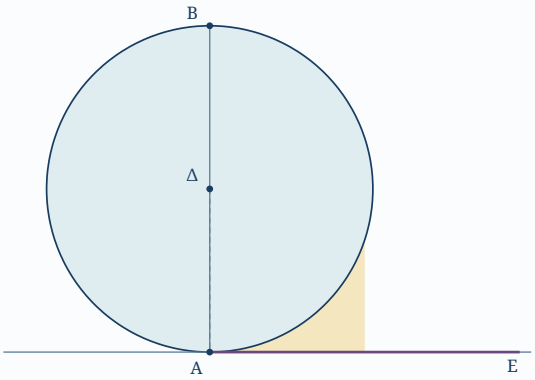
A line that will not fit.

Rise 1, run 1000000. Selected witness: the second crossing.

The ray enters the circle. Its perpendicular foot H is strictly inside: $\Delta H^2 = 1000000000000/1000000000001 < 1$. A smaller positive slope moves the crossing closer to A; it does not make the ray fit the gap.

Euclid, Elements III.16, selected Greek and J. L. Heiberg's Latin, 1883 volume I, pp.208–213. The coordinate certificate and magnifier are modern teaching devices. The source calls the curved remainder smaller than every acute rectilinear angle; we do not assign that curved region a positive number of degrees or construct a system of infinitesimal numbers.

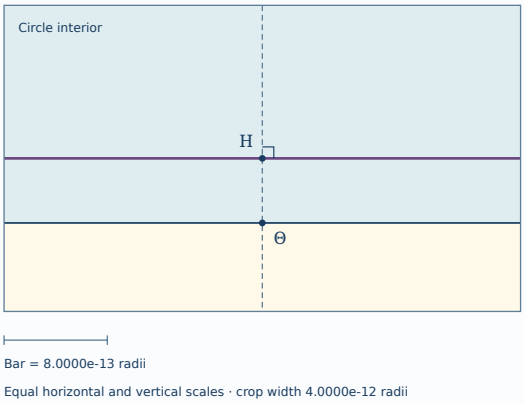
The whole circle



H and K merge with A at this scale. The exact crossing remains positive.

Tan marks part of the curved gap. Purple: the proposed forward ray; a dashed part is its backward extension.

The perpendicular foot, enlarged



Bar = 8.0000e-13 radii

Equal horizontal and vertical scales · crop width 4.0000e-12 radii

Crop width 4.0000e-12 radii; scale bar 8.0000e-13 radii. Horizontal and vertical units are equal. The two plates have different magnifications.

Ray slope: rise / run	1/1000000
H: perpendicular foot (x, y)	(1000000/1000000000001, 1/1000000000001)
ΔH^2	1000000000000/1000000000001

$1 - \Delta H^2$	$1/1000000000001$
K: second crossing (x, y)	$(2000000/1000000000001, 2/1000000000001)$
Positive-ray crossing parameter t	$2/1000000000001$

A point on the line is inside the circle

Put A at (0, 0), Δ at (0, 1), and write the ray as (qt, pt) for $t \geq 0$, with rise p and run q. Its perpendicular foot is $H = (pq/(p^2 + q^2), p^2/(p^2 + q^2))$.

$\Delta H^2 = q^2/(p^2 + q^2)$. If $p > 0$, H lies forward on the ray and $\Delta H^2 < 1$. A point on the ray is inside the circle, so the ray cannot stay in the outside gap. The exact deficit $p^2/(p^2 + q^2)$ remains positive however small a positive p/q is.

Find the other crossing without guessing a pixel

Substitute the ray (qt, pt) into the unit circle $x^2 + (y - 1)^2 = 1$. The difference between squared distance and squared radius is **$t((p^2 + q^2)t - 2p)$**.

Besides $t = 0$ at A, the other root is **$t = 2p/(p^2 + q^2)$**. For $p > 0$ this is positive. Halfway to it lies H, strictly inside. For $p = 0$ both roots coincide at A; for $p < 0$ the other root is on the backward extension. These are different geometric cases.

Why no line fits

The source argues by contradiction

1. First establish that the perpendicular at A lies outside the circle: a second meeting would give an isosceles triangle with two right angles.
2. Suppose a different line through A could stay between that tangent and the circumference. Drop the perpendicular ΔH from the centre to the proposed line.
3. The right triangle gives $\Delta H < \Delta A$, so H is closer to the centre than the circle's radius.
4. But the proposed line was supposed to lie outside: the radius ΔΘ on the way from Δ to H would be shorter than ΔH. Since $\Delta A = \Delta\Theta$, that contradicts step 3.

What the experiment adds

Our realizable diagram puts H where it actually belongs: inside the circle for every positive candidate slope. It does not redraw the source's impossible supposition as though its lengths were consistent.

The coordinate calculation is modern. It confirms the selected rational slope exactly and displays a formula valid for every positive slope. The source's universal claim comes with its geometric argument; a finite collection of tiny lines is not its proof.

The semicircle's curved angle and the remaining curved gap are not assigned degree values. The displayed angle measures only the candidate ray against the tangent. This room does not define addition of curved angles or a system of infinitesimal numbers.

Eight readings, including the bracket

1. Smaller than every rectilinear angle

GREEK · P.208

Ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου, καὶ εἰς τὸν μεταξὺ τόπον τῆς τε εὐθείας καὶ τῆς περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται, καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἁπάσης γωνίας ὀξείας εὐθυγράμμου μείζων ἐστίν, ἡ δὲ λοιπὴ ἐλάττων.

HEIBERG'S LATIN · P.209

Recta, quae ad diametrum circuli in termino perpendicularis erigitur, extra circulum cadet, nec in spatium inter rectam et ambitum ulla alia recta interponetur, et angulus semicirculi quouis acuto angulo rectilineo maior est, reliquus autem minor.

The line drawn at right angles to a circle's diameter from its end will fall outside the circle. No other straight line will fit between that line and the circumference. The angle of the semicircle is greater than every acute rectilinear angle, and the remaining angle is smaller.

The remaining angle has one curved boundary. Rectilinear means bounded by straight lines. The horn angle is a later descriptive name; this statement calls it the remaining angle.

ἁπάσης / quouis

Every / any: the universal comparison carries the claim. Trying many slopes cannot replace it.

εὐθυγράμμου / rectilineo

Straight-lined. Greek genitive singular agrees with γωνίας; Latin ablative gives the object of comparison after maior/minor.

2. Why the perpendicular stays outside

GREEK · P.208

Ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΔΓ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ΔΑΓ γωνίᾳ τῇ ὑπὸ ΑΓΔ. ὀρθὴ δὲ ἡ ὑπὸ ΔΑΓ· ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΑΓΔ· τριγώνου δὴ τοῦ ΑΓΔ αἱ δύο γωνίαι αἱ ὑπὸ ΔΑΓ, ΑΓΔ δύο ὀρθαῖς ἴσαι εἰσίν· ὅπερ ἐστὶν ἀδύνατον.

HEIBERG'S LATIN · P.209

quoniam ΔA = ΔΓ, erit etiam ∠ ΔΑΓ = ΑΓΔ. uerum ∠ ΔΑΓ rectus est. itaque etiam ∠ ΑΓΔ rectus. ergo trianguli ΑΓΔ duo anguli ΔΑΓ + ΑΓΔ duobus rectis aequales sunt; quod fieri non potest.

Since ΔA equals ΔΓ, angle ΔΑΓ also equals angle ΑΓΔ. But ΔΑΓ is right, so ΑΓΔ is right too. Thus two angles of triangle ΑΓΔ equal two right angles, which is impossible.

This is the first contradiction, before the gap argument: suppose the perpendicular met the circumference again at Γ. Equal radii give equal base angles (I.5), whereas any two triangle angles together are less than two right angles (I.17). Γ belongs to that hypothetical diagram; K in our experiment is a separately added crossing of a tilted ray.

ἴση … τῇ / aequales

Equal to: the Greek dative names the other magnitude. The repeated equality propagates a right angle to the other vertex.

ἀδύνατον / fieri non potest

Impossible / cannot happen. The proof rejects its temporary supposition, rather than accepting a triangle with two right angles.

3. Try to put a line into the gap

GREEK · P.210

Πιπτέτω ὡς ἡ ΑΕ· λέγω δή, ὅτι εἰς τὸν μεταξὺ τόπον τῆς τε ΑΕ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται.

HEIBERG'S LATIN · P.211

cadat ut AE. dico, in spatium inter rectam AE et ambitum ΓΘA aliam rectam interponi non posse.

Let it fall as AE. I now say that no other straight line can be put into the space between straight line AE and circumference ΓΘA.

The ray must start at A on the circle's side of the tangent. A ray directed below the tangent stays outside, but it is on the wrong side to occupy this gap. A coincident tangent supplies no positive rectilinear angle.

μεταξὺ τόπον / spatium inter

The space between: τόπον is accusative after εἰς. Both boundaries matter; merely staying outside the circle is insufficient.

οὐ παρεμπεσεῖται / interponi non posse

Will not fall between / cannot be interposed. The Greek future denies the proposed construction; the Latin uses an infinitive after dico.

4. Drop the perpendicular

GREEK · P.210

Εἰ γὰρ δυνατόν, παρεμπιπτέτω ὡς ἡ ΖΑ, καὶ ἤχθω ἀπὸ τοῦ Δ σημείου ἐπὶ τὴν ΖΑ κάθετος ἡ ΔΗ. καὶ ἐπεὶ ὀρθή ἐστιν ἡ ὑπὸ ΑΗΔ, ἐλάττων δὲ ὀρθῆς ἡ ὑπὸ ΔΑΗ, μείζων ἄρα ἡ ΑΔ τῆς ΔΗ. ἴση δὲ ἡ ΔΑ τῇ ΔΘ· μείζων ἄρα ἡ ΔΘ τῆς ΔΗ, ἡ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον.

HEIBERG'S LATIN · P.211

nam, si fieri potest, interponatur ut ZA, et a Δ puncto ad ZA perpendicularis ducatur ΔH. et quoniam ∠ AHΔ rectus est, et ∠ ΔAH minor recto, erit AΔ > ΔH. sed ΔA = ΔΘ. ergo ΔΘ > ΔH, minor maiore; quod fieri non potest.

For, if possible, let a line be put between them as ZA, and draw ΔH perpendicular from Δ to ZA. Since AHΔ is right and ΔAH is less than right, AΔ is greater than ΔH. But ΔA equals ΔΘ. So ΔΘ is greater than ΔH—the smaller greater than the larger—which is impossible.

The contradiction uses the temporary assumption that the line lies beyond the circumference: along ΔH, the circle would be reached at Θ before H. The right triangle instead puts H closer to Δ than a radius. Our realizable diagram shows H inside the circle; it does not pretend to realize the impossible outside arrangement. The print's Latin reference is to I.19.

ἤχθω / ducatur

Let it be drawn: Greek perfect passive imperative, Latin present passive subjunctive. Both issue a construction instruction.

ἐλάττων … μείζων / minor … maiore

Smaller / greater. The opposed comparatives expose the contradiction about the same two distances.

5. One right angle, two curved parts

GREEK · P.210

Λέγω, ὅτι καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἡ περιεχομένη ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἁπάσης γωνίας ὀξείας εὐθυγράμμου μείζων ἐστίν, ἡ δὲ λοιπὴ ἡ περιεχομένη ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας ἁπάσης γωνίας ὀξείας εὐθυγράμμου ἐλάττων ἐστίν.

HEIBERG'S LATIN · P.211

dico etiam, angulum semicirculi recta BA et arcu ΓΘA comprehensum quouis acuto angulo rectilineo maiorem esse, reliquum autem arcu ΓΘA et recta AE comprehensum quouis acuto angulo rectilineo minorem esse.

I also say that the semicircle's angle, bounded by straight line BA and circumference ΓΘA, exceeds every acute rectilinear angle; the remaining angle, bounded by circumference ΓΘA and straight line AE, is less than every acute rectilinear angle.

Do not confuse the ‘angle of the semicircle’ here with an inscribed angle in a semicircle. This one has a curved side. The two curved parts divide the right-angle region at A. Our displayed degree value always belongs to the candidate ray and the tangent, never to the curved remainder.

ἡ μὲν … ἡ δὲ / … autem

The one … the other: the paired articles organize two different comparisons, not two numerical degree measurements.

περιεχομένη / comprehensum

Bounded / enclosed. The following straight line and circumference specify the angle’s two sides.

6. The missing line decides both comparisons

GREEK · P.212

οὐ παρεμπίπτει δέ· οὐκ ἄρα τῆς περιεχομένης γωνίας ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἔσται μείζων ὀξεῖα ὑπὸ εὐθειῶν περιεχομένη, οὐδὲ μὴν ἐλάττων τῆς περιεχομένης ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας.

HEIBERG’S LATIN · P.213

uerum non interponitur recta. itaque nullus angulus acutus rectis comprehensus maior erit angulo comprehenso recta BA et arcu ΓΘA nec minor angulo comprehenso arcu ΓΘA et recta AE.

But no line falls between them. Therefore no acute angle bounded by straight lines will be greater than the angle bounded by BA and circumference ΓΘA, nor smaller than the angle bounded by circumference ΓΘA and straight line AE.

This selected conclusion follows the intervening reductio across pp.210–213; the whole printed argument is available in the source viewer. Euclid’s geometric comparison is the historical argument. Our algebra supplies an additional exact witness for every permitted coordinate trial.

οὐκ ἄρα / itaque nullus

Therefore not / therefore none: the conclusion depends on the excluded line, rather than the number of examples tried.

οὐδὲ μήν / nec

Nor indeed: the second denial completes the paired angle comparisons.

7. A corollary about touching

GREEK · P.212

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ’ ἄκρας ἀγομένη ἐφάπτεται τοῦ κύκλου.

HEIBERG'S LATIN · P.213

Hinc manifestum est, rectam ad diametrum circuli in termino perpendicularem erectam circulum contingere.

From this it is clear that the straight line drawn at right angles to a circle's diameter from its end touches the circle.

The source heading is Πόρισμα / Corollarium. This selection gives the common unbracketed claim. The next reading preserves a further Greek addition, printed in square brackets, for which Heiberg supplies an editorial discussion rather than a facing Latin translation.

φανερόν / manifestum

Clear / evident. The implication comes from the preceding geometric argument.

ἐφάπτεται / contingere

Touches. Greek ἐφάπτεται takes the genitive τοῦ κύκλου; Latin contingere takes the accusative circulum.

8. A bracket is evidence too

GREEK · P.212

[καὶ ὅτι εὐθεῖα κύκλου καθ' ἓν μόνον ἐφάπτεται σημεῖον, ἐπειδήπερ καὶ ἡ κατὰ δύο αὐτῷ συμβάλλουσα ἐντὸς αὐτοῦ πίπτουσα ἐδείχθη]

HEIBERG'S EDITORIAL NOTE · NOT A TRANSLATION · P.213

Pars altera corollarii, per se quoque suspecta, sine dubio a Theone addita est; om. praeter P m. 1 etiam Campanus.

[And that a straight line touches a circle at only one point, since a line meeting it at two points was also shown to fall inside it.]

The Greek brackets belong to the edition. The Latin is the opening of Heiberg's footnote, not a parallel translation: he judges the second part of the corollary an addition by Theon and reports its omission in witnesses. His attribution is an editorial judgment, not a date established by our experiment. The complete footnote remains on p.213.

καθ' ἓν μόνον / Pars altera corollarii

At only one [point]. The Latin instead says 'the second part of the corollary': it discusses the Greek addition rather than translating it.

ἐδείχθη / a Theone addita est

Was shown: Greek aorist passive refers back to a proof. The Latin 'was added by Theon' is Heiberg's separate claim about transmission.

The complete printed witness

Heiberg, 1883, volume I, printed pp.208–213. PDF pages 222–227. Complete pages reduced without cropping or retouching. The source diagram depicts the hypothetical line; our realizable trial is reflected and adds K. The bracketed corollary and its Latin editorial note remain distinct.

Complete public-domain source PDF

ἡ *MN* τῇ *BΓ* ἐδείχθη ἴση [καὶ ἡ *BΓ* τῆς *ZH* μείζων ἐστίν]. μεγίστη μὲν ἄρα ἡ *AΔ* διάμετρος, μείζων δὲ ἡ *BΓ* τῆς *ZH*.

Ἐν κύκλῳ ἄρα μεγίστη μέν ἐστιν ἡ διάμετρος, τῶν δὲ ἄλλων ἀεὶ ἡ ἔγγιον τοῦ κέντρου τῆς ἀπώτερον μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.

ις'.

Ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου, καὶ εἰς τὸν μεταξὺ τόπον τῆς τε εὐθείας καὶ τῆς περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται, καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἁπάσης γωνίας ὀξείας εὐθυγράμμου μείζων ἐστίν, ἡ δὲ λοιπὴ ἐλάττων.

Ἔστω κύκλος ὁ *ABΓ* περὶ κέντρον τὸ *Δ* καὶ διάμετρον τὴν *AB*· λέγω, ὅτι ἡ ἀπὸ τοῦ *A* τῇ *AB* πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐκτὸς πεσεῖται τοῦ κύκλου.

Μὴ γάρ, ἀλλ' εἰ δυνατόν, πιπτέτω ἐντὸς ὡς ἡ *ΓA*, καὶ ἐπεζεύχθω η *ΔΓ*.

Ἐπεὶ ἴση ἐστὶν ἡ *ΔA* τῇ *ΔΓ*, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ *ΔAΓ* γωνίᾳ τῇ ὑπὸ *AΓΔ*. ὀρθὴ δὲ ἡ ὑπὸ *ΔAΓ*· ὀρθὴ ἄρα καὶ ἡ ὑπὸ *AΓΔ*· τριγώνου δὴ τοῦ *AΓΔ* αἱ δύο γωνίαι αἱ ὑπὸ *ΔAΓ*, *AΓΔ* δύο ὀρθαῖς ἴσαι εἰσίν· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ

XVI. Eutocius in Apollonium p. 44. 59.

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1. ἐδείχθη] in ras. V. *BΓ*] *ΓB* B; *BΓ* ἄρα p. 2. ἐστί BV. μέν] m. 2 V. 4. δέ] δ' BF. 5. αἰεί FV. ἔγγειον P, sed corr. τοῦ κέντρου] τῆς διαμέτρου P. 7. ις'] ιη' F; corr. m. 2. 9. ἀγομένη εὐθεῖα F et B m. rec.

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$MN = B\Gamma$. itaque maxima est diametrus $A\Delta$, et

$$B\Gamma > ZH.$$

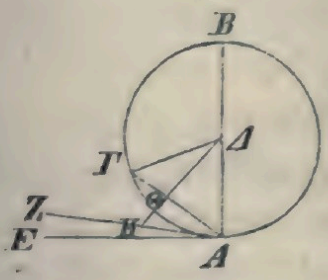
Ergo in circulo maxima est diametrus, ceterarum autem proxima quaeque centro remotiore maior est; quod erat demonstrandum.

XVI.

Recta, quae ad diametrum circuli in termino perpendicularis erigitur, extra circulum cadet, nec in spatium inter rectam et ambitum ulla alia recta interponetur, et angulus semicirculi quouis acuto angulo rectilineo maior est, reliquus autem minor.

Sit circulus $AB\Gamma$ circum centrum Δ et diametrum AB descriptus. dico, rectam ad AB in A termino perpendicularem erectam extra circulum cadere.

ne cadat enim, sed, si fieri potest, intra cadat ut $A\Gamma$, et ducatur $\Delta\Gamma$. quoniam $\Delta A = \Delta\Gamma$, erit etiam $\angle \Delta A\Gamma = A\Gamma\Delta$ [I,5]. uerum $\angle \Delta A\Gamma$ rectus est. itaque etiam $\angle A\Gamma\Delta$ rectus. ergo trianguli $A\Gamma\Delta$ duo anguli $\Delta A\Gamma + A\Gamma\Delta$ duobus rectis aequales sunt; quod fieri non potest [I, 17]. itaque recta ad BA in



12. *πάσης* B. 13. *ἐστίν*] *ἔσται* in ras. V. 16. *AB*] (prius) inter *A* et *B* 1 litt. eras. in V. 19. *ὡς*] supra m. 2 F. *AΓ* p. 21. *ἐπεί*] *ἐπεὶ οὖν* p, ante *ἐπεί* add. *καί* m. 2 FV. *ἴση ἐστί*] om. P. *γωνία*] om. BVp. 22. *AΓΔ ἐστιν ἴση* P. 23. *ΔAΓ*] *Δ* eras. p. *ἄρα*] om. B. *ἡ*] supra m. 1 F. *τριγώνου δὴ τοῦ AΓΔ αἱ δύο γωνίαι αἱ*] P (*AΓ* pro *AΓΔ*); *αἱ ἄρα* Theon? (BFVp; *ἄρα* et seq. *ὑπό* supra m. 2 F). 24. *δυσίν* V. 25. *εἰσιν ἴσαι* B. *ἐστίν*] om. p. *τοῦ*] om. V.

Printed p.209 · PDF p.223 · SHA-256
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Α σημείου τῇ ΒΑ πρὸς ὀρθὰς ἀγομένη ἐντὸς πεσεῖται τοῦ κύκλου. ὁμοίως δὴ δείξομεν, ὅτι οὐδ' ἐπὶ τῆς περιφερείας· ἐκτὸς ἄρα.

Πιπτέτω ὡς ἡ ΑΕ· λέγω δή, ὅτι εἰς τὸν μεταξὺ τόπον τῆς τε ΑΕ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἑτέρα εὐθεῖα οὐ παρεμπεσεῖται.

Εἰ γὰρ δυνατόν, παρεμπιπτέτω ὡς ἡ ΖΑ, καὶ ἤχθω ἀπὸ τοῦ Δ σημείου ἐπὶ τὴν ΖΑ κάθετος ἡ ΔΗ. καὶ ἐπεὶ ὀρθή ἐστιν ἡ ὑπὸ ΑΗΔ, ἐλάττων δὲ ὀρθῆς ἡ ὑπὸ ΔΑΗ, μείζων ἄρα ἡ ΑΔ τῆς ΔΗ. ἴση δὲ ἡ ΔΑ τῇ ΔΘ· μείζων ἄρα ἡ ΔΘ τῆς ΔΗ, ἡ ἐλάττων τῆς μείζονος· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα εἰς τὸν μεταξὺ τόπον τῆς τε εὐθείας καὶ τῆς περιφερείας ἑτέρα εὐθεῖα παρεμπεσεῖται.

Λέγω, ὅτι καὶ ἡ μὲν τοῦ ἡμικυκλίου γωνία ἡ περιεχομένη ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἁπάσης γωνίας ὀξείας εὐθυγράμμου μείζων ἐστίν, ἡ δὲ λοιπὴ ἡ περιεχομένη ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας ἁπάσης γωνίας ὀξείας εὐθυγράμμου ἐλάττων ἐστίν.

Εἰ γὰρ ἐστί τις γωνία εὐθύγραμμος μείζων μὲν τῆς περιεχομένης ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας, ἐλάττων δὲ τῆς περιεχομένης ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας, εἰς τὸν μεταξὺ τόπον τῆς τε ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας εὐθεῖα παρεμπεσεῖται, ἥτις ποιήσει μείζονα μὲν τῆς περιεχομένης ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ὑπὸ εὐθειῶν περιεχομένην,

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1. *ἀπ' ἄκρας ἀγομένη* p. 2. *οὐδέ* BFp. 4. *δή*] om. V. 4. *ΓΘΑ*] corr. ex *ΓΒΑ* m. 2 V. 6. *οὐκ ἐμπεσεῖται* F; *παρ-* add. m. 2. 7. *παρεπιπτέτω*, add. *μ* m. 1, F. *ἡ*]

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A puncto perpendicularis erecta intra circulum non cadet. similiter demonstrabimus, eam ne in ambitum quidem cadere. extra igitur cadet.

cadat ut AE. dico, in spatium inter rectam AE et ambitum $\Gamma\Theta A$ aliam rectam interponi non posse.

nam, si fieri potest, interponatur ut ZA, et a Δ puncto ad ZA perpendicularis ducatur ΔH. et quoniam $\angle AH\Delta$ rectus est, et $\angle \Delta AH$ minor recto, erit $A\Delta > \Delta H$ [I, 19]. sed $\Delta A = \Delta\Theta$. ergo $\Delta\Theta > \Delta H$, minor maiore; quod fieri non potest. itaque in spatium inter rectam et ambitum positum alia recta non interponetur.

dico etiam, angulum semicirculi recta BA et arcu $\Gamma\Theta A$ comprehensum quouis acuto angulo rectilineo maiorem esse, reliquum autem arcu $\Gamma\Theta A$ et recta AE comprehensum quouis acuto angulo rectilineo minorem esse.

nam si quis erit angulus rectilineus angulo comprehenso recta BA et arcu $\Gamma\Theta A$ maior, et idem minor angulo comprehenso arcu $\Gamma\Theta A$ et recta AE, in spatium inter arcum $\Gamma\Theta A$ et rectam AE positum recta interponetur, quae angulum efficiat rectis comprehensum maiorem angulo comprehenso recta BA et arcu $\Gamma\Theta A$ et alium minorem angulo comprehenso arcu

---

in ras. m. 2 V. 9. ἐλάσσων p. 10. ΔA] $A\Delta$ P. 11. τῇ] τῆς φ. $\Delta\Theta$] Θ in ras. p. ἄρα] ἄρα καί p. ἐλάσσων pφ. 12. ἐστίν] om. Bp. 13. τε] om. V. 16. τε] om. BVp. $\Gamma\Theta A$] Γ om. B; m. 2 V. 17. ὀξείας γωνίας p. 18. ἡ] (alt.) om. P, m. rec. B. τε] om. Bp. 19. ὀξείας γωνίας p. ὀξείας] om. B; m. 2 V. 21. ἐστιν P. τις] om. p; m. rec. B. 22. τε] om. p. BA] AB p. 23. ἐλάσσων F. 24. τε τῆς] om. B; τῆς p. 25. τόπον] supra m. 1 P. 26. εὐθεῖα] om. p; m. rec. B. εὐθεῖα, ἥτις p. 28. ὑπό] τὴν ὑπό B, ὑπό τε F (τε eras.). ὑπὸ εὐθειῶν περιεχομένην] om. p. περιεχομένην] -ν m. 2 V; περιελομένην P.

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ἐλάττονα δὲ τῆς περιεχομένης ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας. οὐ παρεμπίπτει δέ· οὐκ ἄρα τῆς περιεχομένης γωνίας ὑπό τε τῆς ΒΑ εὐθείας καὶ τῆς ΓΘΑ περιφερείας ἔσται μείζων ὀξεῖα ὑπὸ εὐθειῶν περιεχομένη, οὐδὲ μὴν ἐλάττων τῆς περιεχομένης ὑπό τε τῆς ΓΘΑ περιφερείας καὶ τῆς ΑΕ εὐθείας.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῇ διαμέτρῳ τοῦ κύκλου πρὸς ὀρθὰς ἀπ' ἄκρας ἀγομένη ἐφάπτεται τοῦ κύκλου [καὶ ὅτι εὐθεῖα κύκλου καθ' ἓν μόνον ἐφάπτεται σημεῖον, ἐπειδήπερ καὶ ἡ κατὰ δύο αὐτῷ συμβάλλουσα ἐντὸς αὐτοῦ πίπτουσα ἐδείχθη]· ὅπερ ἔδει δεῖξαι.

ιξ'.

Ἀπὸ τοῦ δοθέντος σημείου τοῦ δοθέντος κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

Ἔστω τὸ μὲν δοθὲν σημεῖον τὸ Α, ὁ δὲ δοθεὶς κύκλος ὁ ΒΓΔ· δεῖ δὴ ἀπὸ τοῦ Α σημείου τοῦ ΒΓΔ κύκλου ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ε, καὶ ἐπεζεύχθω ἡ ΑΕ, καὶ κέντρῳ μὲν τῷ Ε διαστήματι δὲ τῷ ΕΑ κύκλος γεγράφθω ὁ ΑΖΗ, καὶ ἀπὸ τοῦ

XVI. πόρισμα. Simplicius in phys. fol. 12ᵛ.

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1. ἐλάσσονα p. τε] m. 2 V. 3. τε] om. Bp. 5. ἡ ὑπό V m. 2. οὐ μὴν οὐδέ F. 6. τε] om. p. 8. πόρισμα] comp. Bp, V m. 2; om. PF, V m. 1. 9. τούτων p. ἡ] supra m. 1 P. 11. καὶ ὅτι — 14. δεῖξαι] mg. m. rec. P. 12.

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$\Gamma\Theta A$ et recta AE. uerum non interponitur recta [u. supra]. itaque nullus angulus acutus rectis comprehensus maior erit angulo comprehenso recta BA et arcu $\Gamma\Theta A$ nec minor angulo comprehenso arcu $\Gamma\Theta A$ et recta AE.

Corollarium.

Hinc manifestum est, rectam ad diametrum circuli in termino perpendicularem erectam circulum contingere [def. 2].[1]) — quod erat demonstrandum.

XVII.

A dato puncto datum circulum contingentem rectam lineam ducere.

Sit datum punctum A, datus autem circulus $B\Gamma\Delta$. oportet igitur a puncto A circulum $B\Gamma\Delta$ contingentem rectam lineam ducere.

sumatur enim centrum circuli E, et ducatur AE, et centro E radio autem EA describatur circulus AZH,

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1) Pars altera corollarii, per se quoque suspecta, sine dubio a Theone addita est; om. praeter P m. 1 etiam Campanus. et re uera corollarium genuinum eodem redit. itaque e uerbis Simplicii concludi nequit, eum partem alteram legisse.

ἅπτεται FV. 13. *ὅπερ ἔδει δεῖξαι*] postea insert. F. 15. *ιζ΄*] *ιϑ΄* F; corr. m. 2. 18. *ἔστω* — 20. *ἀγαγεῖν*] *εἰλήφθω γὰρ τοῦ δοθέντος κύκλου τοῦ BΓΔ τὸ δοθὲν σημεῖον τὸ A, καὶ ἔστω τὸ κέντρον τοῦ κύκλου τὸ E.* V; in mg. m. 2: *ἐν ἄλλῳ οὕτως γράφεται· ἔστω τὸ μὲν δοθὲν σημεῖον τὸ A ὁ δὲ δοθεὶς κύκλος ὁ BΓΔ· δεῖ δὴ ἀπὸ δοθέντος σημείου τοῦ A τοῦ δοθέντος κύκλου τοῦ BΓΔ ἐφαπτομένην εὐθεῖαν γραμμὴν ἀγαγεῖν*, et ita B, et p (*ἀπὸ τοῦ δοθέντος*). 19. *A*] om. *φ*. 21. *εἰλήφθω — τὸ E*] mg. m. 2 V. 22. *κέντρον φ*. 23. *EA*] P in ras. m. 1; F; *AE* BVp.

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Keep the editable record


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Reopen this experiment in the published workshop

The reading contains no scripts or remote subresources. Its native data links work without the app. The app link needs the published workshop.


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